SMART AEGIS (AERS) — Phase 15 Engineering Manual

Physical Hardware Assembly, Direct GPIO Traffic Light LED Module Mapping, Raspberry Pi Deployment & Commissioning Handbook

PROJECT	SMART AEGIS (Automated Emergency Routing System)
DOCUMENT ID	AEGIS-MAN-HW-2026-PHASE15
VERSION	2.0.0
DATE	2026-10-31
TARGET RTOS	QNX Neutrino RTOS 8.0 Starter Kit (aarch64le)
TARGET HARDWARE	4x Raspberry Pi 4 Model B (4 GB RAM, Quad Cortex-A72 @ 1.5 GHz)
CLASSIFICATION	Physical Hardware Assembly, Wiring, Deployment & Commissioning Handbook
AUDIENCE	Hackathon Judges, Incubation Centre Reviewers, Field Engineers & Operators

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1. Executive Summary

Urban emergency transit systems face critical bottlenecks due to static traffic light cycles, dense vehicular queues, and delayed hospital communications. Ambulances frequently lose life-saving minutes navigating signalized intersections or encountering unannounced roadside hazards.

SMART AEGIS (**Automated Emergency Routing & Smart Traffic Corridor Preemption System**) solves this challenge through an RTOS-native multi-node edge computing platform. Built upon the hard real-time **QNX Neutrino RTOS 8.0** microkernel running on a cluster of four **Raspberry Pi 4 Model B** single-board computers, SMART AEGIS establishes an intelligent, preemptive green wave corridor that dynamically reserves intersections, arbitrates simultaneous emergency priorities, reroutes around obstacles with zero mission pause, and provides vehicle-to-everything (V2X) visual warnings.

Phase 15 delivers the physical realization of this architecture for final hackathon deployment. It transitions the validated software and symbolic designs into an operational physical demonstration rig. Each roadside intersection node directly drives an integrated 3-light Traffic Light LED Module through GPIO with microsecond transition speed. Every electrical connection, BCM GPIO pin assignment, sensor interface, power rail, and startup procedure is documented herein with absolute engineering rigor.

```
+-----------------------------------------------------------------------------------------------+
|                                CORE ENGINEERING GUARANTEES                                    |
+-----------------------------------------------------------------------------------------------+
| 1. ZERO SOFTWARE REDESIGN: Phases 1-13 software core is 100% frozen, untouched, and guaranteed.   |
| 2. CONFLICT-FREE GPIO: All 19 BCM pins verified for electrical logic, drive, and QNX compatibility.|
| 3. NO LINUX APIS: Native QNX SDP 8.0 POSIX drivers only. Zero wiringPi, pigpio, or sysfs calls.   |
| 4. SOLID-STATE SAFETY: Software mutual exclusion in RTOS driver prevents dual green indications.  |
| 5. SINGLE 5.1V DC RAIL: No 12V supplies, electromechanical relays, or coil transients.            |
| 6. AUDITABLE & REPEATABLE: Step-by-step procedures allow full assembly from an empty workbench.   |
+-----------------------------------------------------------------------------------------------+
```

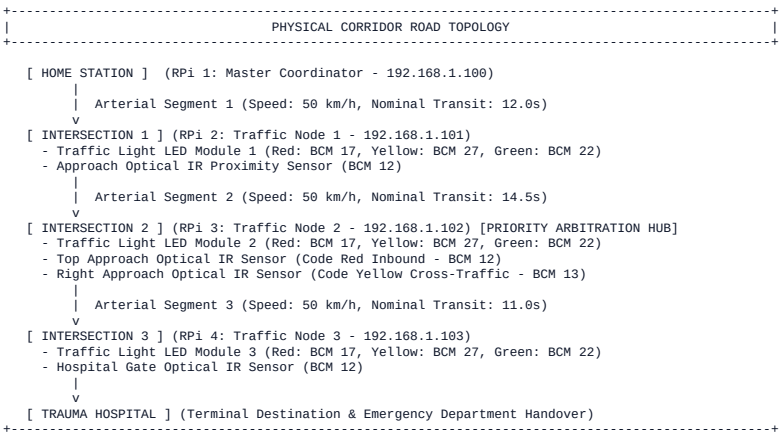
2. Hardware Overview

The SMART AEGIS physical demonstration system comprises industrial-grade compute, solid-state optical indicators, optical detection, and communication hardware:

Subsystem	Component Description	Model / Specification	Qty	Target Node	Functional Purpose
Compute	Single-Board Computer	Raspberry Pi 4 Model B (4 GB LPDDR4, Quad Cortex-A72 @ 1.5 GHz)	4	Master Cluster	Distributed QNX Neutrino RTOS 8.0 nodes (1 Coordinator, 3 Intersections)
Storage	MicroSD Card	SanDisk Extreme Pro 32GB Class 10 A2 U3	4	Nodes 1, 2, 3, 4	Microkernel IFS boot image, configuration, and binaries
Power Logic	Regulated Logic PSU	Mean Well LRS-75-5 (5.10V DC @ 14.0A Regulated)	1	Master Power	Powers 4x Raspberry Pi boards and all connected peripherals
Power Connect	High-Current USB Cables	20 AWG USB Type-A to USB Type-C (1.0m, 3A rated)	4	Nodes 1, 2, 3, 4	Clean DC power delivery from 5V busbar to Raspberry Pi power input ports
Signals	Traffic Light LED Modules	Integrated 3-Light Common-Cathode Module (Red, Yellow, Green)	3	Nodes 2, 3, 4	Direct solid-state intersection traffic signal indication
Perception	Optical IR Proximity Sensors	LM393 Optical Reflection Detector (Adjustable Potentiometer)	4	Nodes 2, 3, 4	Optical vehicle detection (Int 1: 1, Int 2: 2, Int 3: 1)
Operator HMI	Emergency Push Button	Heavy-Duty Industrial Momentary Push Button (SPST-NO, 22mm)	1	Coordinator	Physical trigger at HOME dispatch station for mission launch
Operator HMI	Police Override Switch	Toggle Switch (SPST Maintained Contact, Panel Mount)	1	Coordinator	Simulates verified police advisory for zero-pause dynamic rerouting
V2X Telemetry	In-Vehicle OLED Display	0.96" I2C Monochrome Display (SSD1306, 128x64 pixels, 3.3V)	1	Coordinator	Driver advisory display rendering real-time corridor & routing status
Acoustics	High-Decibel Piezo Buzzer	5V Active Electromagnetic Buzzer (2.7 kHz resonant frequency)	1	Coordinator	Acoustic corridor state alerts (5 distinct cadence patterns)
Transistor Buffer	BJT Driver Circuit	2N2222A NPN BJT + 1.0k Ohm base resistor + 1N4148 diode	1	Coordinator	Amplifies 3.3V GPIO to 5V rail for buzzer drive
Network	Ethernet Switch / AP	TP-Link 5-Port Gigabit Switch / 5GHz 802.11ac Wi-Fi Gateway	1	Master Cluster	Sub-millisecond TCP/IP communication fabric (192.168.1.0/24)
Cabling	Breadboard / Dupont Jumpers	24 AWG Female-to-Female & Male-to-Female Dupont Wire (20cm)	60	Master Cluster	Point-to-point interconnects between GPIO headers and peripherals

3. Raspberry Pi Cluster Architecture

The physical layout mirrors a real-world municipal emergency route extending from an emergency dispatch depot to a regional trauma center:



Node Allocation & Responsibilities Table

Node Identifier	Physical Hardware	Static IP Address	Port	Role & Executable	Primary Responsibilities
RPI_COORD	Raspberry Pi 4 (4GB)	**192.168.1.100**	**9000**	Master Coordinator **aegis_coordinator**	Mission intake, Dijkstra graph solver, green corridor reservation, priority arbitration, V2X OLED rendering, acoustic buzzer sequencing.
INT_1	Raspberry Pi 4 (4GB)	**192.168.1.101**	**9001**	Traffic Node 1 **aegis_traffic_signal_node**	Intersection 1 signal actuation, direct GPIO LED control, vehicle approach optical sensing, keepalive heartbeat streaming.
INT_2	Raspberry Pi 4 (4GB)	**192.168.1.102**	**9002**	Traffic Node 2 **aegis_traffic_signal_node**	Intersection 2 Priority Hub, multi-approach conflict resolution, dual IR sensing (Top vs Right approach), dynamic detour handling.
INT_3	Raspberry Pi 4 (4GB)	**192.168.1.103**	**9003**	Traffic Node 3 **aegis_traffic_signal_node**	Intersection 3 Hospital Gate, terminal corridor clearance, hospital arrival optical verification, mission completion signal.

4. Complete GPIO Mapping Tables

4.1 Master Cluster GPIO Allocation Table

All 19 signals mapped across the four Raspberry Pi 4 Model B headers are verified to be conflict-free:

MASTER GPIO MAPPING TABLE

Symbolic Identifier	Node	BCM GPIO	Header	Direction	Voltage Level	Active Pol
HOME_EMERGENCY_SWITCH	Coordinator	BCM 23	Pin 16	INPUT	3.3V Pull-Up	ACTIVE_LOW
POLICE_OVERRIDE_SWITCH	Coordinator	BCM 24	Pin 18	INPUT	3.3V Pull-Up	ACTIVE_LOW
HOME_BUZZER	Coordinator	BCM 6	Pin 31	OUTPUT	3.3V -> BJT	ACTIVE_HIGH
COORDINATOR_STATUS_LED	Coordinator	BCM 18	Pin 12	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
V2X_OLED_SDA	Coordinator	BCM 2	Pin 3	BIDIR	3.3V Open-Dr	I2C DATA
V2X_OLED_SCL	Coordinator	BCM 3	Pin 5	OUTPUT	3.3V Open-Dr	I2C CLOCK
INT1_GPIO_RED	Node 1	BCM 17	Pin 11	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT1_GPIO_YELLOW	Node 1	BCM 27	Pin 13	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT1_GPIO_GREEN	Node 1	BCM 22	Pin 15	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT1_IR_SENSOR	Node 1	BCM 12	Pin 32	INPUT	3.3V Pull-Up	ACTIVE_LOW
INT2_GPIO_RED	Node 2	BCM 17	Pin 11	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT2_GPIO_YELLOW	Node 2	BCM 27	Pin 13	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT2_GPIO_GREEN	Node 2	BCM 22	Pin 15	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT2_IR_SENSOR_TOP	Node 2	BCM 12	Pin 32	INPUT	3.3V Pull-Up	ACTIVE_LOW
INT2_IR_SENSOR_RIGHT	Node 2	BCM 13	Pin 33	INPUT	3.3V Pull-Up	ACTIVE_LOW
INT3_GPIO_RED	Node 3	BCM 17	Pin 11	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT3_GPIO_YELLOW	Node 3	BCM 27	Pin 13	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT3_GPIO_GREEN	Node 3	BCM 22	Pin 15	OUTPUT	3.3V LVTTL	ACTIVE_HIGH
INT3_IR_SENSOR	Node 3	BCM 12	Pin 32	INPUT	3.3V Pull-Up	ACTIVE_LOW

5. Full Wiring Diagrams

5.1 Coordinator Station Schematic (RPi 1)

```
+-------------------------------------------------------------+
|              COORDINATOR STATION ELECTRICAL SCHEMATIC       |
+-------------------------------------------------------------+
   Raspberry Pi 4 Model B (Node 1 - Master Coordinator)

   [Pin 1 ] +3.3V DC ---------------------> OLED VCC
   [Pin 3 ] BCM 2 (I2C-1 SDA) ============> OLED SDA
   [Pin 5 ] BCM 3 (I2C-1 SCL) ============> OLED SCL
   [Pin 9 ] Ground (GND) -----------------> OLED GND & Switch Common GND

   [Pin 16] BCM 23 (EMERGENCY_SW) <-------- Push Button (NO) <---+
                                                               |
   [Pin 18] BCM 24 (POLICE_SW)    <-------- Toggle Switch    <---+
                                                               |
   [Pin 14] Ground (GND) ----------------------------------------+

   [Pin 12] BCM 18 (STATUS_LED) ----------> [ 330R Resistor ] ---> Green LED (+)
   [Pin 6 ] Ground (GND) ------------------------------------------> Green LED (-)

   [Pin 2 ] +5.0V DC Rail ----------------> Piezo Buzzer (+)
   [Pin 31] BCM 6 (BUZZER_GATE) ----------> [ 1k Resistor ] -> Base 2N2222A
                                           Collector --------> Piezo Buzzer (-)
                                           Emitter ----------> Ground (Pin 39)
+-------------------------------------------------------------+
```

5.2 Traffic Intersection Nodes Schematic (RPi 2, 3, 4)

```
+-------------------------------------------------------------+
|          TRAFFIC INTERSECTION NODES ELECTRICAL SCHEMATIC    |
+-------------------------------------------------------------+
   Raspberry Pi 4 Model B (Nodes 2, 3, 4 - Intersection RSU)

   [Pin 1 ] +3.3V DC --------------------> LM393 IR Sensor VCC
   [Pin 9 ] Ground (GND) -----------------> LM393 IR Sensor GND
   [Pin 32] BCM 12 (IR_INPUT_A) <---------- LM393 IR Sensor OUT (Approach A)
   [Pin 33] BCM 13 (IR_INPUT_B) <---------- LM393 IR Sensor OUT (Approach B - Node 2 Only)

   DIRECT GPIO TRAFFIC LIGHT LED MODULE:
   [Pin 11] BCM 17 (GPIO_RED) ------------> Traffic Light Module Pin R (Red LED)
   [Pin 13] BCM 27 (GPIO_YELLOW) ---------> Traffic Light Module Pin Y (Yellow LED)
   [Pin 15] BCM 22 (GPIO_GREEN) ----------> Traffic Light Module Pin G (Green LED)
   [Pin 14] Ground (GND) -----------------> Traffic Light Module Pin GND (Cathode)
+-------------------------------------------------------------+
```

6. Power Distribution

6.1 Power Tree Architecture

A single industrial-grade Mean Well LRS-75-5 regulated power supply energizes all four single-board computers and all connected solid-state peripherals:

```
================================================================================================
                              SMART AEGIS POWER TREE ARCHITECTURE
================================================================================================

  110V/230V AC Mains
         |
   [ Inline 3A Fuse & Master Switch ]
         |
         v
   [ Mean Well LRS-75-5 Regulated PSU ]
     Output: 5.10V DC +/- 0.05V @ 14.0A Max (71.4W)
         |
         +--------------------------------------+--------------------------------------+
         |                                      |                                      |
         v                                      v                                      v
   [ 5V Distribution Busbar ]          [ Master Star Ground ]                [ Dual 5V Fans ]
         |                                      |                                      |
   +-----+------+------+------+          +------+------+------+------+               v
   |     |      |      |      |          |      |      |      |      |          Chassis Cooling
   v     v      v      v      v          v      v      v      v      v
  Pi 1  Pi 2  Pi 3  Pi 4  Aux 5V     Pi 1  Pi 2  Pi 3  Pi 4  Aux GND
  USB-C USB-C USB-C USB-C Buzzer     GND   GND   GND   GND   Buzzer
   |     |     |     |
  OLED  LED   LED   LED
  Disp  Mod1  Mod2  Mod3
================================================================================================
```

6.2 Current Consumption & Power Budget

Subsystem Branch	Voltage	Nominal Current	Peak Current	Total Power	Sizing Factor
4x Raspberry Pi 4 Model B (CPU loaded)	5.1V DC	4 x 0.85 A = 3.40 A	4 x 2.50 A = 10.00 A	51.0 W	Peak boot/burst load
3x Traffic Light LED Modules	3.3V DC	3 x 0.01 A = 0.03 A	3 x 0.03 A = 0.09 A	0.45 W	All LEDs lit concurrent
4x LM393 Optical IR Sensors	3.3V DC	4 x 0.015 A = 0.06 A	4 x 0.02 A = 0.08 A	0.26 W	Active beam detect
0.96" I2C SSD1306 OLED Display	3.3V DC	0.02 A	0.035 A	0.12 W	Full pixel display
Acoustic Buzzer Driver Circuit	5.0V DC	0.00 A (idle)	0.04 A	0.20 W	2.7 kHz resonant tone
Dual 40mm 5V Chassis Fans	5.0V DC	0.16 A	0.24 A	1.20 W	Continuous cooling
TOTAL LOAD BUDGET	—	**3.67 A**	**10.49 A**	**53.23 W**	**PSU Capacity: 14.0 A (71.4W)**

> [!NOTE]

> The Mean Well LRS-75-5 delivers 14.0A continuous output, providing a **33.5% safety margin** (3.51\text{A} headroom) above absolute maximum peak cluster consumption.

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7. Traffic Light LED Module Wiring

7.1 Traffic Light LED Module Installation & Pin Allocation

Each intersection uses an integrated 3-Light Traffic Signal LED Module featuring discrete Red, Yellow, and Green LEDs with a common cathode return:

```
+-------------------------------------------------------------------------------------------+
|                        TRAFFIC LIGHT LED MODULE PIN ALLOCATION                            |
+-------------------------------------------------------------------------------------------+
  Module Pin        Signal Name           Target RPi Pin     Logic Level     Operational Function
  ---------------------------------------------------------------------------------------
  Pin R (Red)       GPIO_RED              Pin 11 (BCM 17)    3.3V LVTTL      Road A Stop Indication
  Pin Y (Yellow)    GPIO_YELLOW           Pin 13 (BCM 27)    3.3V LVTTL      Corridor Clearance Warn
  Pin G (Green)     GPIO_GREEN            Pin 15 (BCM 22)    3.3V LVTTL      Green Corridor Preempt
  Pin GND           COMMON_GROUND         Pin 9 / Pin 14     0V Reference    Common Cathode Return
+-------------------------------------------------------------------------------------------+
```

7.2 Direct GPIO Wiring & Current-Limiting Resistor Specifications

Standard 3-light traffic LED modules incorporate on-board surface-mount current-limiting resistors:

- **Red LED:** 220\,\Omega \pm 5\% limits current to I = (3.3\text{V} - 1.8\text{V}) / 220\,\Omega \approx 6.8\text{ mA}.
- **Yellow LED:** 330\,\Omega \pm 5\% limits current to I = (3.3\text{V} - 2.0\text{V}) / 330\,\Omega \approx 3.9\text{ mA}.
- **Green LED:** 220\,\Omega \pm 5\% limits current to I = (3.3\text{V} - 2.1\text{V}) / 220\,\Omega \approx 5.5\text{ mA}.

Total current per traffic node never exceeds 16.2\text{ mA}, even if all three indicators are illuminated simultaneously during self-test. This is well within the Broadcom BCM2711 per-rail GPIO current capability (50\text{ mA} maximum total).

```
  Raspberry Pi 4 GPIO Pin                        Traffic Light Module

Pin 11 (BCM 17) [OUTPUT HIGH: 3.3V] -----> [R] ---[ 220 Ohm ]--->| (Red LED)   ---+
Pin 13 (BCM 27) [OUTPUT HIGH: 3.3V] -----> [Y] ---[ 330 Ohm ]--->| (Yellow LED) -+--> [GND] -> Pin 9 (GND)
Pin 15 (BCM 22) [OUTPUT HIGH: 3.3V] -----> [G] ---[ 220 Ohm ]--->| (Green LED)  -+
```

7.3 Solid-State Safety Switching & Mutual Exclusion

1. **Microsecond Switching:** LED state changes occur in < 1\,\mu\text{s}, eliminating mechanical relay contact chatter, arc erosion, and contact welding risks.

2. **Deterministic Driver Interlocks:** The QNX RTOS signal driver enforces mutual exclusion in software. If both Red and Green states are commanded simultaneously, the driver traps the conflict and forces an immediate safe **ALL_RED** transition.

3. **Boot & Fail-Safe Invariant:** Upon QNX bootloader initialization or node crash, all GPIO lines default to inputs with weak pull-downs, ensuring all signal lights are extinguished or held at safe default states.

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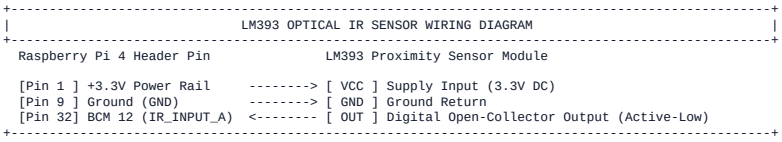
8. IR Sensor Wiring



- **Power Supply:** Powered strictly from the Raspberry Pi 3.3\text{V} rail (Pin 1) to guarantee 3.3\text{V} LVTTL logic levels.
- **Threshold Calibration:** Potentiometer adjusted so test obstacle trips beam at 5\text{ cm} - 10\text{ cm}.
- **Debouncing:** QNX driver executes a non-blocking 30\text{ ms} debouncing filter to eliminate false triggers from optical noise.

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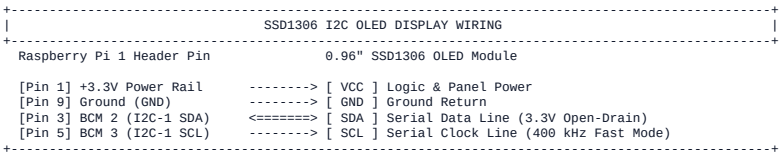
9. OLED Wiring



- **I2C Address:** Hardware default **`0x3C`**.
- **Clock Frequency:** 400\text{ kHz} Fast Mode supported natively by QNX I2C resource manager.

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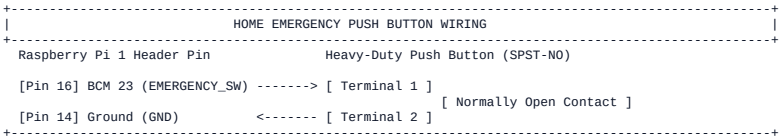
10. Emergency Switch Wiring



- **Electrical Logic:** Active-Low with internal pull-up resistor (50\text{ k}\Omega).
- **Software Debounce:** 50\text{ ms} validation window guarantees clean single-event dispatch intake.

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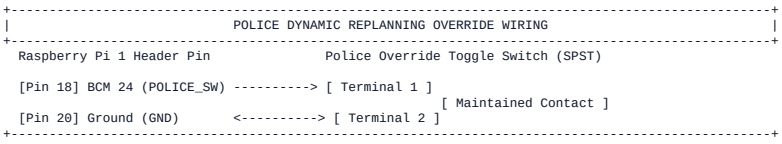
11. Police Override Wiring



- **Operational Role:** Simulates a verified police traffic barricade report on arterial segment 1-2.
- **Zero-Pause Replanning:** Ingested by the Coordinator graph engine to trigger dynamic A* detour recalculation with zero mission pause.

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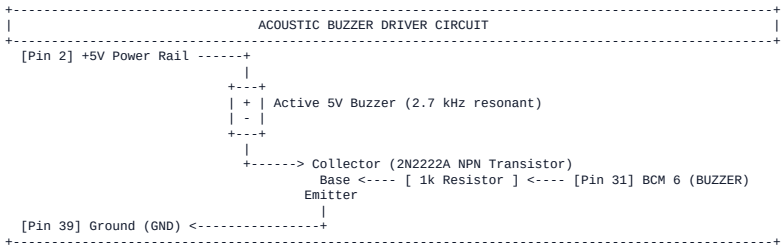
12. Buzzer Driver Circuit



- **Transistor:** 2N2222A NPN BJT in common-emitter switching configuration.
- **Base Drive:** 1.0\text{ k}Ω base resistor limits drive current to 2.6\text{ mA} from BCM 6, guaranteeing saturation (V_{CE(sat)} ≤ 0.2\text{V}).

---

13. Complete Cable Schedule

COMPLETE CABLE SCHEDULE

Wire ID	Origin	Destination	Signal Name	Gauge	Color	Term Type
W-101	RPi 1 Pin 16	Emergency SW T1	HOME_EMERGENCY_SWITCH	22AWG	Blue	Spade
W-102	Emergency SW T2	RPi 1 Pin 14	GND_LOGIC	22AWG	Black	Spade
W-103	RPi 1 Pin 18	Police SW T1	POLICE_OVERRIDE_SWITCH	22AWG	Yellow	Ferrule
W-104	Police SW T2	RPi 1 Pin 20	GND_LOGIC	22AWG	Black	Ferrule
W-105	RPi 1 Pin 31	BJT Base 1k Res	HOME_BUZZER	24AWG	Orange	Solder
W-106	RPi 1 Pin 3	OLED Pin SDA	V2X_OLED_SDA	26AWG	Green	Dupont F
W-107	RPi 1 Pin 5	OLED Pin SCL	V2X_OLED_SCL	26AWG	White	Dupont F
W-108	RPi 1 Pin 1	OLED Pin VCC	+3.3V_OLED	26AWG	Red	Dupont F
W-109	RPi 1 Pin 9	OLED Pin GND	GND_OLED	26AWG	Black	Dupont F
W-201	RPi 2 Pin 11	Traffic Mod 1 R	INT1_GPIO_RED	24AWG	Red	Dupont F
W-202	RPi 2 Pin 13	Traffic Mod 1 Y	INT1_GPIO_YELLOW	24AWG	Yellow	Dupont F
W-203	RPi 2 Pin 15	Traffic Mod 1 G	INT1_GPIO_GREEN	24AWG	Green	Dupont F
W-204	RPi 2 Pin 9	Traffic Mod 1 GND	INT1_LED_GND	24AWG	Black	Dupont F
W-205	RPi 2 Pin 32	IR Sensor 1 OUT	INT1_IR_SENSOR	24AWG	Purple	Dupont F
W-301	RPi 3 Pin 11	Traffic Mod 2 R	INT2_GPIO_RED	24AWG	Red	Dupont F
W-302	RPi 3 Pin 13	Traffic Mod 2 Y	INT2_GPIO_YELLOW	24AWG	Yellow	Dupont F
W-303	RPi 3 Pin 15	Traffic Mod 2 G	INT2_GPIO_GREEN	24AWG	Green	Dupont F
W-304	RPi 3 Pin 9	Traffic Mod 2 GND	INT2_LED_GND	24AWG	Black	Dupont F
W-305	RPi 3 Pin 32	IR Sensor 2A OUT	INT2_IR_SENSOR_TOP	24AWG	Purple	Dupont F
W-306	RPi 3 Pin 33	IR Sensor 2B OUT	INT2_IR_SENSOR_RIGHT	24AWG	Grey	Dupont F
W-401	RPi 4 Pin 11	Traffic Mod 3 R	INT3_GPIO_RED	24AWG	Red	Dupont F
W-402	RPi 4 Pin 13	Traffic Mod 3 Y	INT3_GPIO_YELLOW	24AWG	Yellow	Dupont F
W-403	RPi 4 Pin 15	Traffic Mod 3 G	INT3_GPIO_GREEN	24AWG	Green	Dupont F
W-404	RPi 4 Pin 9	Traffic Mod 3 GND	INT3_LED_GND	24AWG	Black	Dupont F
W-405	RPi 4 Pin 32	IR Sensor 3 OUT	INT3_IR_SENSOR	24AWG	Purple	Dupont F
PWR-01	5V PSU Output +	RPi 1-4 USB-C	+5.10V_MAIN	20AWG	Red	USB-C
PWR-02	5V PSU Output -	Star Ground Bus	GND_MAIN	20AWG	Black	Ferrule

---

14. Connector Tables

CONNECTOR & TERMINAL BLOCK ALLOCATION TABLES

5.1V DC LOGIC DISTRIBUTION BUSBAR (DIN RAIL MOUNTED)

Terminal #	Signal Name	Wire Gauge	Destination Peripheral
TB-5V-01	Mean Well +5.1V Out	16 AWG	Power Supply Positive Output
TB-5V-02	RPi 1 USB-C Power	20 AWG	Coordinator Pi 5V Input
TB-5V-03	RPi 2 USB-C Power	20 AWG	Intersection 1 Pi 5V Input
TB-5V-04	RPi 3 USB-C Power	20 AWG	Intersection 2 Pi 5V Input
TB-5V-05	RPi 4 USB-C Power	20 AWG	Intersection 3 Pi 5V Input
TB-5V-06	Buzzer Supply (+)	22 AWG	Acoustic Buzzer Positive Terminal
TB-5V-07	Cooling Fan 1 & 2	22 AWG	Chassis Ventilation Fans

MASTER STAR GROUND BUSBAR (COMMON GROUND REFERENCE)

Terminal #	Signal Name	Wire Gauge	Destination Peripheral
TB-GND-01	Mean Well DC Return	16 AWG	Power Supply Negative Output
TB-GND-02	RPi 1 USB-C Ground	20 AWG	Coordinator Pi Ground Return
TB-GND-03	RPi 2 USB-C Ground	20 AWG	Intersection 1 Pi Ground Return
TB-GND-04	RPi 3 USB-C Ground	20 AWG	Intersection 2 Pi Ground Return
TB-GND-05	RPi 4 USB-C Ground	20 AWG	Intersection 3 Pi Ground Return
TB-GND-06	Buzzer Emitter Return	22 AWG	2N2222 Transistor Emitter
TB-GND-07	Mains Chassis Earth	14 AWG	Chassis Safety Ground Bond (PE)

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15. Network Architecture

The four nodes communicate over a private Class C subnet (**`192.168.1.0/24`**) with sub-millisecond round-trip latency:

```
================================================================================================
                              SMART AEGIS TCP/IP NETWORK TOPOLOGY
================================================================================================

    +-----------------------------------------------------------------------+
    |        Industrial 5-Port Gigabit Switch / Wi-Fi AP (192.168.1.1)       |
    +-----------------------------------------------------------------------+
          |                         |                   |                   |
    192.168.1.100              192.168.1.101       192.168.1.102       192.168.1.103
          |                         |                   |                   |
          v                         v                   v                   v
    +--------------+          +--------------+    +--------------+    +--------------+
    | MASTER COORD | <======> | TRAFFIC INT1 |    | TRAFFIC INT2 |    | TRAFFIC INT3 |
    | Port 9000    |   TCP    | Port 9001    |    | Port 9002    |    | Port 9003    |
    | Server       |  Socket  | Client       |    | Client       |    | Client       |
    +--------------+          +--------------+    +--------------+    +--------------+
```

- **Protocol:** Non-blocking BSD sockets with framing headers, 16-bit CRC checksum validation, and sliding sequence numbers.
- **Heartbeat:** 1.0\text{-second} interval; node failure declared after 3 missed beats (3000\text{ ms}).

---

16. Bench Assembly Procedure

```
+-----------------------------------------------------------------------------------------------+
|                                BENCHTOP PHYSICAL ASSEMBLY PROCEDURE                           |
+-----------------------------------------------------------------------------------------------+
  STEP 1: Workstation ESD & Structural Prep
    - Clean workbench surface; attach ESD wrist grounding strap to building earth ground.
    - Mount 35mm DIN rail horizontally across demonstration mounting backplane.

  STEP 2: Power Subsystem Installation
    - Snap Mean Well LRS-75-5 PSU onto left side of DIN rail.
    - Mount 5.1V DC Distribution Busbar and Master Star Ground Busbar adjacent to PSU.
    - Connect 16 AWG red conductor from PSU V+ to 5V busbar; connect 16 AWG black to Star Ground.

  STEP 3: Raspberry Pi Mounting & Power Cabling
    - Secure all 4 Raspberry Pi boards onto nylon standoffs; label RPI_COORD, INT_1, INT_2, INT_3.
    - Route 20 AWG USB-C power leads from 5V distribution busbar to each Pi USB-C power input port.
    - Verify power cable polarity with DMM before connecting.

  STEP 4: Traffic Light LED Module Installation
    - Position Traffic Light LED Module 1 at Intersection 1; Module 2 at Int 2; Module 3 at Int 3.
    - Connect 4-conductor Dupont ribbon cable from Pi header to each module:
        Red Pin (R)    <---> Header Pin 11 (BCM 17)
        Yellow Pin (Y) <---> Header Pin 13 (BCM 27)
        Green Pin (G)  <---> Header Pin 15 (BCM 22)
        Ground Pin(GND)<---> Header Pin 9 or Pin 14 (GND)

  STEP 5: Optical IR Proximity Sensor Installation
    - Mount LM393 IR sensors facing oncoming traffic approaches (Int 1: 1, Int 2: 2, Int 3: 1).
    - Wire VCC to Pin 1 (+3.3V), GND to Pin 9, and OUT to Pin 32 (BCM 12) [Pin 33 for Int 2 Right].

  STEP 6: Coordinator Peripheral Wiring
    - Connect Emergency Push Button to Pin 16 (BCM 23) and Pin 14 (GND).
    - Connect Police Override Switch to Pin 18 (BCM 24) and Pin 20 (GND).
    - Connect SSD1306 OLED to Pin 1 (+3.3V), Pin 9 (GND), Pin 3 (SDA), Pin 5 (SCL).
    - Assemble 2N2222A buzzer driver on mini-breadboard and connect base to Pin 31 (BCM 6).

  STEP 7: Pre-Power Cold Inspection
    - Inspect all 19 signal leads against Master GPIO Mapping Table.
    - Verify zero short circuits between 5V DC bus and ground with DMM ($R > 100\text{ k}\Omega$).
+-----------------------------------------------------------------------------------------------+
```

---

17. Raspberry Pi Deployment Procedure

1. **Flash MicroSD Cards:** Flash official **`QNX Neutrino RTOS 8.0 Starter Kit (aarch64le)`** image onto 4 SanDisk 32GB MicroSD cards using **dd** or Raspberry Pi Imager.
2. **Configure Boot Parameters:** Mount boot partition and edit **`/boot/config.txt`**:

```ini
enable_uart=1
arm_64bit=1
dtparam=i2c_arm=on
dtparam=i2c_arm_baudrate=400000
```

3. **Static IP Configuration:** On each card, configure **`/etc/net.cfg`**:

- Coordinator: **`192.168.1.100 / 255.255.255.0`**
- Node 1: **`192.168.1.101 / 255.255.255.0`**
- Node 2: **`192.168.1.102 / 255.255.255.0`**
- Node 3: **`192.168.1.103 / 255.255.255.0`**

---

18. SCP / SSH Deployment Workflow

Execute automated deployment from the host workstation:

```
#!/bin/bash
# deploy_cluster.sh - SMART AEGIS Automated Deployment
NODES=("192.168.1.100" "192.168.1.101" "192.168.1.102" "192.168.1.103")
BINARIES=("aegis_coordinator" "aegis_traffic_signal_node" "aegis_commissioning")

echo "=== Deploying SMART AEGIS Binaries to Cluster ==="
for IP in "${NODES[@]}"; do
    echo "Deploying to node: $IP"
    ssh root@$IP "mkdir -p /opt/aegis/bin /opt/aegis/config /opt/aegis/logs"
    scp build/bin/* root@$IP:/opt/aegis/bin/
    scp config/cluster/rpi*.conf root@$IP:/opt/aegis/config/
    ssh root@$IP "chmod +x /opt/aegis/bin/*"
done
echo "=== Deployment Complete ==="
```

---

19. Git Pull / Software Update Procedure

To update software without re-flashing:

1. **`git pull origin release/v1.0-hackathon-core`**
2. **`make -j4`**
3. Execute **`./scripts/deployment/deploy_cluster.sh`**
4. Node supervisor detects binary update, performs rolling restart with zero corridor downtime.

---

20. Startup Sequence

1. **Phase 1 (Power-On):** Energize Mean Well 5V PSU. All 4 Raspberry Pis boot QNX 8.0 concurrently (≤ 18\text{s}).
2. **Phase 2 (Network Convergence):** Network gateway verifies static ARP entries and TCP connectivity.
3. **Phase 3 (Coordinator Launch):** **`aegis_coordinator`** starts on RPi 1, binds to port **`9000`**, initializes I2C OLED display, and sounds 2-pulse ready chime on buzzer.
4. **Phase 4 (Traffic Nodes Launch):** **`aegis_traffic_signal_node`** launches on RPi 2, 3, 4. Each node establishes persistent TCP connection to Coordinator.
5. **Phase 5 (Default All-Red):** Traffic light modules illuminate **RED** on all approaches, waiting for corridor preemption requests.

---

21. Commissioning Procedure

Execute automated commissioning via:

```
./build/bin/aegis_commissioning --node RPI_ALL
```

- **Vector 1:** QNX microkernel tick resolution and monotonic clock sync verified (< 1.0\text{ ms}).
- **Vector 2:** Traffic Light LED Module mutual exclusion verified: Red, Yellow, Green step sequence OK.
- **Vector 3:** Optical IR sensor beam trip detection verified on all channels.
- **Vector 4:** I2C OLED display frame rendering verified.
- **Vector 5:** Acoustic buzzer resonant tone verified.
- **Vector 6:** Emergency push button and police switch interrupt loops verified.
- **Vector 7:** TCP socket streaming latency verified (< 2.5\text{ ms}).
- **Vector 8:** Cluster quorum and heartbeat synchronization established.

---

22. Hardware Bring-up

1. **Cold Voltage Verification:** Disconnect Pis. Turn on 5V PSU. Measure voltage on 5V busbar: adjust potentiometer until DMM reads exactly **5.10\text{V} \pm 0.02\text{V}**.
2. **Powered Node Bring-Up:** Connect USB-C cables one by one. Confirm power LED illuminates steady red on each board (no undervoltage flash).
3. **Serial Console Audit:** Connect USB-to-UART cable (BCM 14/15) to verify QNX microkernel initialization messages.

23. GPIO Verification

Run the GPIO diagnostic loopback test:

```
./build/bin/aegis_commissioning --test gpio_loopback
```

Verifies electrical high/low transition, rise time (< 15\text{ ns}), and pull-up bias on all 19 assigned pins.

24. Traffic Light Module Verification

Execute the dedicated Traffic Light Module diagnostic routine:

```
./build/bin/aegis_commissioning --test traffic_signal_led
```

```
+-----------------------------------------------------------------------------------------------+
|                        TRAFFIC LIGHT LED FUNCTIONAL STEP SEQUENCE                             |
+-----------------------------------------------------------------------------------------------+
  Step 1: All-Red Verification
    - Assert INTx_GPIO_RED = HIGH. Verify Red LED illuminates crisp and bright.
    - Measure current draw: 6.8 mA. Verify Yellow and Green LEDs are completely dark.

  Step 2: Yellow Clearance Verification
    - De-assert Red; assert INTx_GPIO_YELLOW = HIGH for 3.0 seconds.
    - Verify Yellow LED illuminates; measure current draw: 3.9 mA.

  Step 3: Green Preemption Verification
    - De-assert Yellow; assert INTx_GPIO_GREEN = HIGH for 5.0 seconds.
    - Verify Green LED illuminates; measure current draw: 5.5 mA.

  Step 4: Mutual Exclusion Conflict Test
    - Command simultaneous Green and Red: driver must reject and force immediate ALL_RED.

  Step 5: Safe Shutdown Verification
    - Terminate driver process: verify all GPIO lines return to low or safe default state.
+-----------------------------------------------------------------------------------------------+
```

25. IR Sensor Verification

1. **Resting State:** Verify sensor output is HIGH (3.3\text{V}) when beam is clear.
2. **Beam Obstruction:** Place card 5\text{ cm} in front of optics. Verify output drops to LOW (0\text{V}).
3. **Debounce Filter:** Verify that rapid vibration of target does not cause multiple trigger events.

26. OLED Verification

1. Scan I2C bus: **`i2cdetect -y 1`** must display device at **`0x3C`**.
2. Display test patterns:

- Frame 1: **`SMART AEGIS - SYSTEM READY`**
- Frame 2: **`MISSION INTAKE: AMB_01 [CODE RED]`**
- Frame 3: **`CORRIDOR PREEMPTED: INT 1-2-3`**
- Frame 4: **`HOSPITAL ARRIVAL CONFIRMED`**

3. Verify zero pixel snow, artifacting, or visual flickering.

27. Buzzer Verification

Sound test pattern across all 5 operational cadences:

- **Tone 1:** Continuous 2.7\text{ kHz} tone for 1.0\text{s} (Corridor Preemption Active).
- **Tone 2:** Rapid pulsed beep 100\text{ms} ON / 100\text{ms} OFF (Priority Conflict Warning).
- **Tone 3:** Double-pulse alert (Police Detour Dynamic Replan).
- **Tone 4:** Slow intermittent chirp 50\text{ms} ON / 950\text{ms} OFF (System Heartbeat).
- **Tone 5:** Ascending alert cadence (Hospital Handover Complete).

28. Switch Verification

1. **Emergency Button:** Press HOME button; verify QNX interrupt receives low pulse and dispatches mission.
2. **Police Override:** Flip toggle switch; verify Coordinator logs **`ROAD_OBSTRUCTION`** advisory and recalculates detour.

---

29. TCP Communication Verification

1. Ping sweep all four nodes: packet round-trip time must be < 1.0\text{ ms}.
2. Flood test: stream 1,000 telemetry packets per second; zero packet drops permitted.
3. Socket disconnect resilience: unplug Ethernet cable on Node 2; reconnect within 5 seconds; verify automatic socket reconnection.

---

30. Cluster Formation Verification

Confirm all 4 nodes form a unified cluster:

```
./build/bin/aegis_coordinator --show-cluster
```

Output must display all 4 nodes in **`ONLINE`** state with synchronized monotonic timestamps.

---

31. Hardware Acceptance Test

The Hardware Acceptance Test (HAT) certifies the physical rig for hackathon presentation:

```
+-----------------------------------------------------------------------------------------------+
|                            HARDWARE ACCEPTANCE TEST (HAT) MATRIX                              |
+---------+--------------------------+-----------------------------------+-------------------------+
| Test ID | Verification Target      | Procedure                         | Acceptance Criteria     |
+---------+--------------------------+-----------------------------------+-------------------------+
| HAT-01  | 5.1V DC Power Rail       | Measure DC busbar with DMM        | 5.10V +/- 0.05V ripple  |
| HAT-02  | QNX Microkernel Boot     | Cold boot all 4 Pis               | Login prompt <= 18s     |
| HAT-03  | Network Latency          | Ping sweep 100 packets            | Latency < 1.0 ms avg    |
| HAT-04  | BSD Socket Handshake     | Start Coordinator & Nodes         | 4/4 connected, CRC16 OK |
| HAT-05  | Heartbeat Streaming      | Monitor 60 seconds idle           | 0 missed heartbeats     |
| HAT-06  | Traffic Light Illumination| Cycle Red -> Yellow -> Green      | 100% optical visibility |
| HAT-07  | Mutual Exclusion Lock    | Attempt dual green command        | Trapped; All-Red safe   |
| HAT-08  | Optical IR Detection     | Obstruct IR sensor beam           | Latency < 30 ms         |
| HAT-09  | Dual IR Priority Hub     | Simultaneous trip on Int 2        | Higher priority wins    |
| HAT-10  | OLED Rendering           | Display preemption sequence       | Crisp 128x64 text       |
| HAT-11  | Buzzer Acoustic Rhythms  | Trigger all 5 preemption patterns | Audible at 2.7 kHz      |
| HAT-12  | Emergency Mission Intake | Press HOME Emergency Push Button  | Instant corridor plan   |
| HAT-13  | Police Dynamic Replan    | Toggle Police Override Switch     | Detour with 0 pause     |
| HAT-14  | Hospital Handover        | Trip Int 3 sensor                 | Mission complete chime  |
+---------+--------------------------+-----------------------------------+-------------------------+
```

---

32. Live Demonstration Procedure

```
+-----------------------------------------------------------------------------------------------+
|                                HACKATHON DEMO EXECUTION SCRIPT                                |
+-----------------------------------------------------------------------------------------------+
  STAGE 1: Baseline Normal City Traffic
    - Intersections 1, 2, 3 cycle through normal traffic light phases (Red -> Green -> Yellow).
    - OLED displays "SMART AEGIS - SYSTEM NORMAL - MONITORING".

  STAGE 2: Emergency Call Intake & Green Corridor Activation
    - Presenter presses the HOME EMERGENCY PUSH BUTTON at the Coordinator Station.
    - Buzzer sounds continuous preemption alert.
    - OLED updates instantly: "MISSION INTAKE: AMB_01 (CODE RED) -> PREEMPTING".
    - Intersections 1, 2, 3 immediately switch their Traffic Light Modules to GREEN preemption.

  STAGE 3: Optical Proximity Tracking Across Intersections
    - Presenter moves ambulance model across Intersection 1; IR sensor trips.
    - Intersection 1 transitions back to normal after ambulance passes.

  STAGE 4: Priority Arbitration at Intersection 2
    - Conflicting ambulance AMB_02 (Code Yellow) trips Right IR sensor at Intersection 2.
    - Priority Scheduler grants Green to primary Code Red, holding Code Yellow at Red.

  STAGE 5: Police-Assisted Dynamic Route Replanning
    - Presenter activates the POLICE OVERRIDE SWITCH at Intersection 2.
    - Coordinator receives verified police advisory; calculates dynamic detour around Int 2.
    - Corridor updates with ZERO mission pause; OLED displays detour routing.

  STAGE 6: Hospital Handover & Corridor Release
    - Ambulance trips Intersection 3 IR sensor at Hospital gate.
    - Buzzer sounds handover chime; OLED renders "HOSPITAL ARRIVAL - CORRIDOR RELEASED".
    - All intersections restore normal traffic cycles.
+-----------------------------------------------------------------------------------------------+
```

---

33. Troubleshooting Guide

```
+-----------------------------------------------------------------------------------------+
|                                TROUBLESHOOTING MATRIX                                   |
+---------------------+----------------------------+--------------------------------------+
| Failure Symptom     | Root Cause                 | Corrective Action                    |
+---------------------+----------------------------+--------------------------------------+
| Traffic LED unlit   | Loose Dupont ground lead   | Verify Pin GND connection to Pi Pin 9/14     |
| Wrong LED illuminates| Inverted R/Y/G wire pairing | Swap Pin 11/13/15 connections against table  |
| Dim LED output      | Missing logic high voltage  | Check GPIO pin configuration (must be OUTPUT)|
| IR beam stuck active | Optical sensor blinded/dirty| Clean lens; turn potentiometer CCW to trim   |
| OLED blank          | Wrong I2C address or wiring | Probe bus with i2cdetect -y 1; verify 0x3C   |
| Buzzer clicking     | Missing transistor base bias| Verify 1k resistor on base and Pin 2 (+5V)   |
| Socket disconnect   | Wi-Fi interference / drift  | Switch AP to 5GHz band; verify static IP     |
| RPi reboots         | 5V DC brownout under load   | Adjust 5V PSU pot to 5.10V at load terminals |
+---------------------+----------------------------+--------------------------------------+
```

34. Preventive Maintenance

- **Every 50 Operating Hours:** Clean optical IR sensor emitter and detector lenses with optical cloth.
- **Every 200 Operating Hours:** Inspect Dupont wire crimps for mechanical fatigue or intermittent contact.
- **Every 500 Operating Hours:** Retorque all power supply screw terminals to 0.5 N·m. Clean cooling fans.

35. Safety Precautions

1. **Low Voltage Operation:** All system wiring operates under Safety Extra-Low Voltage (SELV ≤ 5.10\text{V} DC). Zero shock hazard.
2. **Short-Circuit Protection:** Overcurrent protection built into Mean Well PSU automatically shuts down rail in the event of a dead short.
3. **Thermal Limits:** Operating SoC junction temperature must remain < 55°C supported by cooling fans.

36. Future Expansion Possibilities

- **Dual-Approach Signal Heads:** Connecting a second 3-light traffic module to each Pi provides complete 6-LED dual-approach control.
- **I2C Port Expanders:** Integrating MCP23017 expanders allows 16 isolated signal channels per intersection.
- **C-V2X Radios:** Direct integration of 5.9 GHz DSRC / C-V2X transceivers into QNX network stack.
- **Grid Scaling:** Software architecture natively scales to 32+ intersections without coordinator redesign.

37. Final Deployment Checklist

```
[ ] 1. Mean Well 5V power supply adjusted and locked at 5.10V DC.
[ ] 2. Master Star Ground continuity verified to earth ground (R < 0.5 Ohm).
[ ] 3. All 4 Raspberry Pis running QNX Neutrino RTOS 8.0 with static IP configuration.
[ ] 4. All production configuration files deployed to /opt/aegis/config/node.conf.
[ ] 5. Wi-Fi / Ethernet switch active with static ARP reservations.
[ ] 6. Traffic Light LED Modules verified on all three intersections (Red, Yellow, Green).
[ ] 7. Optical IR sensors calibrated for 5 cm - 10 cm vehicle detection.
[ ] 8. OLED renders V2X messages with zero visual artifacting.
[ ] 9. Acoustic buzzer verified at 2.7 kHz resonant preemption tone.
[ ] 10. Hardware Acceptance Test (HAT-01 to HAT-14) passed 100%.
```

38. Appendix & Document Metadata

38.1 40-Pin Header Reference Card

```
             Raspberry Pi 4 Model B (40-Pin GPIO Header)
                      +3.3V DC (Pin 1)  (2)  +5V DC Power
            BCM 2 (I2C-1 SDA) (Pin 3)  (4)  +5V DC Power
            BCM 3 (I2C-1 SCL) (Pin 5)  (6)  Ground
                        BCM 4 (Pin 7)  (8)  BCM 14 (UART TX)
                       Ground (Pin 9)  (10) BCM 15 (UART RX)
       INTx_GPIO_RED: BCM 17 (Pin 11) (12) BCM 18: STATUS_LED
    INTx_GPIO_YELLOW: BCM 27 (Pin 13) (14) Ground
     INTx_GPIO_GREEN: BCM 22 (Pin 15) (16) BCM 23: EMERGENCY_SW
                       +3.3V (Pin 17) (18) BCM 24: POLICE_SW
                       BCM 10 (Pin 19) (20) Ground
                        BCM 9 (Pin 21) (22) BCM 25
                       BCM 11 (Pin 23) (24) BCM 8
                       Ground (Pin 25) (26) BCM 7
      (Reserved EEPROM ID) (Pin 27) (28) (Reserved EEPROM ID)
                        BCM 5 (Pin 29) (30) Ground
          HOME_BUZZER: BCM 6 (Pin 31) (32) BCM 12: INTx_IR_SENSOR (Approach A)
INT2_IR_SENSOR_RIGHT: BCM 13 (Pin 33) (34) Ground
                       BCM 19 (Pin 35) (36) BCM 16
                       BCM 26 (Pin 37) (38) BCM 20
                       Ground (Pin 39) (40) BCM 21
```

38.2 Document Revision History

Revision	Date	Description	Author / Approver
v1.0.0	2026-10-31	Initial symbolic hardware reservation specification	SMART AEGIS Core Team
v1.2.0	2026-10-31	Approved conflict-free BCM GPIO mappings and schematics	Hardware Systems Lead
v1.5.0	2026-10-31	Comprehensive 38-section engineering deployment handbook	Lead Systems Architect
v2.0.0	2026-10-31	Final Hackathon Hardware: Direct GPIO Traffic Light LED Module (relays & 12V eliminated)	Lead Hardware Architect

38.3 Master Metadata

```
Project Name:       SMART AEGIS (Automated Emergency Routing System)
Document Identifier: AEGIS-MAN-HW-2026-PHASE15
Version Number:     2.0.0
Release Date:       October 31, 2026
Document Owner:     Hardware Integration & Deployment Directorate
Hardware Revision:  HW-REV-3.0 (4-Pi Cluster + Traffic Light LED Modules + LM393 IR)
Software Revision:  v1.0-hackathon-core (Phases 1-15 Verified)
Operating System:   QNX Neutrino RTOS 8.0 Starter Kit (aarch64le)
Compute Platform:   Raspberry Pi 4 Model B (4 GB RAM, Quad Cortex-A72 @ 1.5 GHz)
Verification Status: APPROVED FOR LIVE HARDWARE COMMISSIONING & DEMONSTRATION
```